BTEC Level 3 Applied Science · Unit 1 Chemistry · Answer sheet

Ionic Bonding — Answers

Mark scheme for the companion worksheet. Accept equivalent correct wording. These are original JDScience items, not official Pearson questions.

Numbering matches the student worksheet exactly. Award equivalent scientifically correct wording. Show units on every calculated quantity.

A–E

1. Electrostatic attraction between oppositely charged ions.
2. A metal and a non-metal.
3. Ions are not free to move.
4. Na loses 1 e^- → Na^+. Cl gains that electron → Cl^-. Oppositely charged ions attract.
5. Strong attractions throughout a giant lattice need a lot of energy to overcome.
6. Ions become free to move and carry charge.

7. Na_2O; $MgCl_2$; Al_2O_3.

8. A shift lines up like charges, which repel, so the crystal splits.

9. Molten NaCl: mobile ions. Copper: delocalised electrons.

10. Mg loses $2e^- \rightarrow Mg^{2+}$. O gains $2e^- \rightarrow O^{2-}$. Electrostatic attraction. Giant lattice. [4]

11. Strong ionic attractions → high m.p. Solid ions fixed → no conduction. [4]

12. $Al_2(SO_4)_3$ so +6 balances –6. [2]

13. Mg^{2+} and O^{2-} have higher charges than Na^+ and Cl^-, so attraction is stronger.